

React application that demonstrates various JavaScript concepts

Code

File name: App.tsx

```
import React, { useState } from "react";
import NumberList, { NumberItem } from "../components/NumberList";
import FilterControls from "../components/FilterControls";
import Logger from "../components/Logger";
import HoistingDemo from "../components/HoistingDemo";
import ConstructorDemo from "../components/ConstructorDemo";

const App: React.FC = () => {
  const originalNumbers: NumberItem[] = [
    { value: 1 },
    { value: 2 },
    { value: 3 },
    { value: 4 },
    { value: 5 },
  ];

  const [numbers, setNumbers] = useState<NumberItem[]>(originalNumbers);

  return (
    <div style={{ padding: "20px", fontFamily: "Arial" }}>
      <h2>JSX and JavaScript Concepts Sprint</h2>

      <FilterControls
        originalNumbers={originalNumbers}
        setNumbers={setNumbers}
      />
```

```
    <NumberList numbers={numbers} />
    <Logger numbers={numbers} />
    <HoistingDemo />
    <ConstructorDemo />
  </div>
);
};
export default App;
```

File name: main.tsx

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import "./index.css";
```

```
ReactDOM.createRoot(document.getElementById("root") as HTMLElement).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

File name: index.css

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  background-color: #f4f4f9;
  color: #333;
}
```

```
h2, h3 {  
  color: #2c3e50;  
}
```

```
button {  
  margin: 4px;  
  padding: 6px 12px;  
  border: none;  
  background-color: #007bff;  
  color: white;  
  border-radius: 4px;  
  cursor: pointer;  
}
```

```
button:hover {  
  background-color: #0056b3;  
}
```

File name: NumberList.tsx

```
import React from "react";
```

```
export interface NumberItem {  
  value: number;  
}
```

```
interface NumberListProps {  
  numbers: NumberItem[];  
}
```

```
const NumberList: React.FC<NumberListProps> = ({ numbers }) => {
```

```

return (
  <div>
    <h3>Number List</h3>
    <ul>
      {numbers.map((num, index) => (
        <li key={index}>{num.value}</li>
      ))}
    </ul>
  </div>
);
};
export default NumberList;

```

File name: Logger.tsx

```

import React from "react";
import { NumberItem } from "../NumberList";

interface LoggerProps {
  numbers: NumberItem[];
}

const Logger: React.FC<LoggerProps> = ({ numbers }) => {
  const handleLog = () => {
    numbers.forEach((num) => console.log(num.value));
    alert("Numbers have been logged to the console!");
  };

  return (
    <div>
      <h3>Logger Component</h3>
      <button onClick={handleLog}>Log All Numbers</button>
    </div>
  );
};

```

```
    </div>

  );
};

export default Logger;
```

File name: HoistingDemo.tsx

```
import React from "react";

const HoistingDemo: React.FC = () => {

  function demonstrateHoisting() {

    console.log("Before declaration:", a);

    var a = 10;

    console.log("After declaration:", a);

  }

  return (

    <div>

      <h3>Hoisting Demo</h3>

      <button onClick={demonstrateHoisting}>Run Hoisting Example</button>

    </div>

  );
};

export default HoistingDemo;
```

File name: FilterControls.tsx

```
import React from "react";

import { NumberItem } from "../NumberList";

interface FilterControlsProps {
```

```

originalNumbers: NumberItem[];
setNumbers: React.Dispatch<React.SetStateAction<NumberItem[]>>;
}

const FilterControls: React.FC<FilterControlsProps> = ({
  originalNumbers,
  setNumbers,
}) => {
  const filterEven = () => {
    const evenNumbers = originalNumbers.filter((num) => num.value % 2 === 0);
    setNumbers(evenNumbers);
  };

  const mapDouble = () => {
    const doubled = originalNumbers.map((num) => ({
      value: num.value * 2,
    }));
    setNumbers(doubled);
  };

  const resetList = () => {
    setNumbers(originalNumbers);
  };

  const logNumbers = () => {
    originalNumbers.forEach((num) => console.log(num.value));
    alert("Numbers logged to console!");
  };

  return (
    <div>

```

```

    <h3>Filter & Map Controls</h3>

    <button onClick={filterEven}>Filter Even Numbers</button>{" "}

    <button onClick={mapDouble}>Map (Double Numbers)</button>{" "}

    <button onClick={logNumbers}>Log Numbers</button>{" "}

    <button onClick={resetList}>Reset</button>

  </div>

);

};

export default FilterControls;

```

File name: ConstructorDemo.tsx

```

import React from "react";

class Person {
  name: string;
  age: number;

  constructor(name: string, age: number) {
    this.name = name;
    this.age = age;
  }

  introduce() {
    return `Hi, I'm ${this.name}, and I'm ${this.age} years old.`;
  }
}

const ConstructorDemo: React.FC = () => {
  const person = new Person("Kiran Pavan Pamidi", 22);

  return (

```

```
<div>

  <h3>Constructor Demo</h3>

  <p>{person.introduce()}</p>

</div>

);

};

export default ConstructorDemo;
```

File name: index.html

```
<!doctype html/>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <link rel="icon" type="image/svg+xml" href="/vite.svg" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>my-react-app</title>

  </head>

  <body>

    <div id="root"></div>

    <script type="module" src="/src/main.tsx"></script>

  </body>

</html>
```


Output Screenshots



